

Elementary Linear Algebra

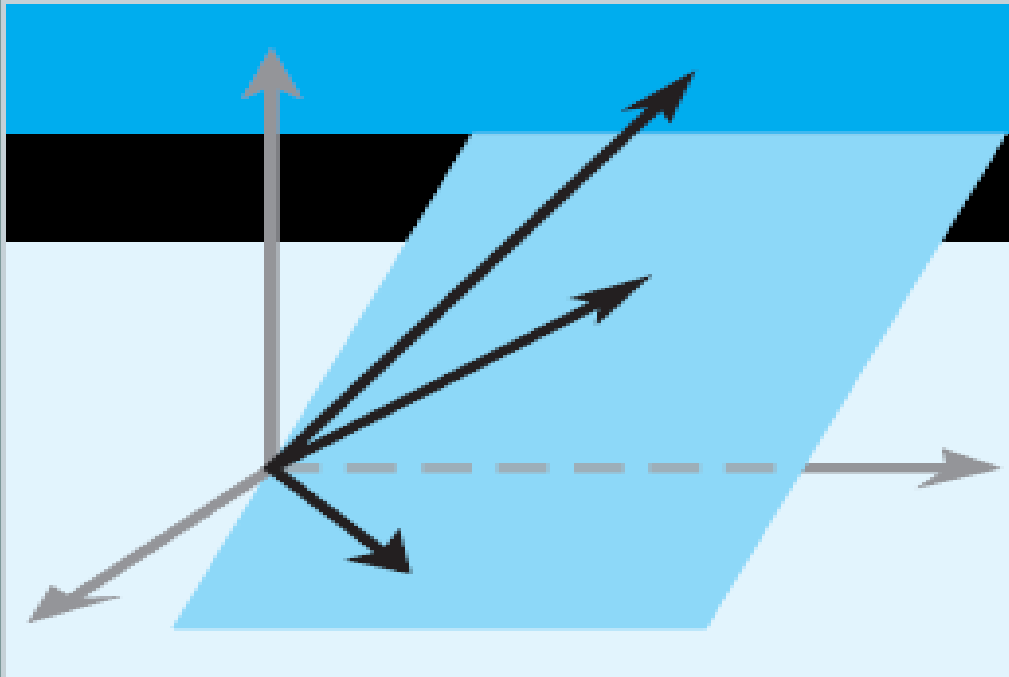


Chapter 5

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Chapter 5 Eigenvalues and Eigenvectors



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Section 5.2 Diagonalization

DEFINITION 1 If A and B are square matrices, then we say that B is similar to A if there is an invertible matrix P such that $B = P^{-1}AP$.

$$\det(B) = ?$$

$$\begin{aligned}\det(B) &= \det(P^{-1}AP) \\ &= \det(APP^{-1}) \\ &= \det(AI) \\ &= \det(A)\end{aligned}$$

by Theorem 2.3.4

$$\det(CD) = \det(C)\det(D)$$

Section 5.2 Diagonalization

Table 1 Similarity Invariants

Property	Description
Determinant	A and $P^{-1}AP$ have the same determinant.
Invertibility	A is invertible if and only if $P^{-1}AP$ is invertible.
Rank	A and $P^{-1}AP$ have the same rank.
Nullity	A and $P^{-1}AP$ have the same nullity.
Trace	A and $P^{-1}AP$ have the same trace.
Characteristic polynomial	A and $P^{-1}AP$ have the same characteristic polynomial.
Eigenvalues	A and $P^{-1}AP$ have the same eigenvalues.
Eigenspace dimension	If λ is an eigenvalue of A and hence of $P^{-1}AP$, then the eigenspace of A corresponding to λ and the eigenspace of $P^{-1}AP$ corresponding to λ have the same dimension.

Section 5.2 Diagonalization

DEFINITION 2 A square matrix A is said to be diagonalizable if it is similar to some diagonal matrix; that is, if there exists an invertible matrix P such that $P^{-1}AP$ is diagonal. In this case the matrix P is said to *diagonalize* A .

$$P^{-1}AP = D$$

$$AP = PD$$

$$D = \begin{bmatrix} d_{11} & 0 & 0 & 0 \\ 0 & d_{22} & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & d_{nn} \end{bmatrix}$$

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THEOREM 5.2.1

If A is an $n \times n$ matrix, the following statements are equivalent.

- (a) A is diagonalizable.*
- (b) A has n linearly independent eigenvectors.*

Proof

Example 7 of Section 5.1

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

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► EXAMPLE 1 Finding a Matrix P That Diagonalizes a Matrix A

Find a matrix P that diagonalizes *Example 7 of Section 5.1*

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

Solution

In Example 7 of the preceding section we found the characteristic equation of A to be

$$(\lambda - 1)(\lambda - 2)^2 = 0$$

and we found the following bases for the eigenspaces:

$$\lambda = 2: \quad \mathbf{p}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{p}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad \lambda = 1: \quad \mathbf{p}_3 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ *linearly independent?*

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Example 1

There are three basis vectors in total, so the matrix

$\lambda=2 \ \lambda=2 \ \lambda=1$

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

diagonalizes A . As a check, you should verify that

$$P^{-1}AP = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Section 5.2 Diagonalization

► EXAMPLE 2 A Matrix That Is Not Diagonalizable

Find a matrix P that diagonalizes

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$$

Solution

The characteristic polynomial of A is

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & 0 & 0 \\ -1 & \lambda - 2 & 0 \\ 3 & -5 & \lambda - 2 \end{vmatrix} = (\lambda - 1)(\lambda - 2)^2$$

so the characteristic equation is

$$(\lambda - 1)(\lambda - 2)^2 = 0$$

Section 5.2 Diagonalization

Example 2

Thus, the distinct eigenvalues of A are $\lambda = 1$ and $\lambda = 2$. We leave it for you to show that bases for the eigenspaces are

$$\lambda = 1: \mathbf{p}_1 = \begin{bmatrix} \frac{1}{8} \\ -\frac{1}{8} \\ 1 \end{bmatrix}; \quad \lambda = 2: \mathbf{p}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Since A is a 3×3 matrix and there are only two basis vectors in total, A is not diagonalizable.



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Procedure for Diagonalizing a Matrix

Step 1. Confirm that the matrix is actually diagonalizable by finding n linearly independent eigenvectors. One way to do this is by finding a basis for each eigenspace and merging these basis vectors into a single set S . If this set has fewer than n vectors, then the matrix is not diagonalizable.

Step 2. Form the matrix $P = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_n]$ that has the vectors in S as its column vectors.

Step 3. The matrix $P^{-1}AP$ will be diagonal and have the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ corresponding to the eigenvectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ as its successive diagonal entries.

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THEOREM 5.2.2 *If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are distinct eigenvectors of a matrix A corresponding to distinct eigenvalues, then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a linearly independent set.*

THEOREM 5.2.3 *If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonaliz-able.*

Section 5.2 Diagonalization

► EXAMPLE 3 Using Theorem 5.2.3

We saw in Example 3 of the preceding section that

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

has three distinct eigenvalues: $\lambda = 4$, $\lambda = 2 + \sqrt{3}$, and $\lambda = 2 - \sqrt{3}$. Therefore, A is diagonalizable and

$$P^{-1}AP = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 + \sqrt{3} & 0 \\ 0 & 0 & 2 - \sqrt{3} \end{bmatrix}$$

for some invertible matrix P . If needed, the matrix P can be found using the method shown in Example 1 of this section.

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► EXAMPLE 4 Diagonalizability of Triangular Matrices

From Theorem [5.1.2](#), the eigenvalues of a triangular matrix are the entries on its main diagonal. Thus, a triangular matrix with distinct entries on the main diagonal is diagonalizable. For example,

$$A = \begin{bmatrix} -1 & 2 & 4 & 0 \\ 0 & 3 & 1 & 7 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

is a diagonalizable matrix with eigenvalues $\lambda_1 = -1$, $\lambda_2 = 3$, $\lambda_3 = 5$, $\lambda_4 = -2$.

by Theorem 5.2.3

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$$P^{-1}AP = D$$

$$D = \begin{bmatrix} d_{11} & 0 & 0 & 0 \\ 0 & d_{22} & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & d_{nn} \end{bmatrix}$$

$$\Rightarrow A = PDP^{-1}$$

$$\Rightarrow A^k = (PDP^{-1})^k$$

$$\Rightarrow A^2 = (PDP^{-1})^2$$

$$= PDP^{-1}PDP^{-1} = PD^2P^{-1}$$

$$\Rightarrow A^k = PD^kP^{-1} \quad \underline{3}$$

Section 5.2 Diagonalization

▶ EXAMPLE 5 Powers of a Matrix

Use [3](#) to find A^{13} , where

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

Solution

We showed in Example [1](#) that the matrix A is diagonalized by

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

and that

$$D = P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Example 5

Thus, it follows from [3](#) that

$$\begin{aligned} A^{13} = P D^{13} P^{-1} &= \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2^{13} & 0 & 0 \\ 0 & 2^{13} & 0 \\ 0 & 0 & 1^{13} \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \quad (4) \\ &= \begin{bmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{bmatrix} \end{aligned}$$



THEOREM 5.2.4

If λ is an eigenvalue of a square matrix $\mathbf{A}$ and $\mathbf{x}$ is a corresponding eigenvector, and if k is any positive integer, then λ^k is an eigenvalue of $\mathbf{A}^k$ and $\mathbf{x}$ is a corresponding eigenvector.

Theorem 5.1.4

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►EXAMPLE 6 The Converse of Theorem 5.2.3 Is False

Consider the matrices

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

It follows from Theorem 5.1.2 that both of these matrices have only one distinct eigen-value, namely $\lambda = 1$, and hence only one eigenspace. We leave it as an exercise for you to solve the characteristic equations

$$(\lambda I - I) \mathbf{x} = \mathbf{0} \quad \text{and} \quad (\lambda I - J) \mathbf{x} = \mathbf{0}$$

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Example 6

with $\lambda = 1$ and show that for I the eigenspace is three-dimensional (all of $\mathbf{R}^3$) and for J it is one-dimensional, consisting of all scalar multiples of

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This shows that the converse of Theorem 5.2.3 is false, since we have produced two 3×3 matrices with fewer than three distinct eigenvalues, one of which is diagonalizable and the other of which is not.

Section 5.2 Diagonalization

Definition

If λ_0 is an eigenvalue of an $n \times n$ matrix A , then the dimension of the eigenspace corresponding to λ_0 is called *geometric multiplicity* of λ_0

$(\lambda - 1)(\lambda - 2)^2 = 0$ *Example 7 of Section 5.1*

A basis for the eigenspace corresponding to $\lambda = 2$.

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

A basis for the eigenspace corresponding to $\lambda = 1$.

$$\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

geometric multiplicity of $\lambda = 1$ is

geometric multiplicity of $\lambda = 2$ is

Section 5.2 Diagonalization

Definition

If λ_0 is an eigenvalue of an $n \times n$ matrix A , then the dimension of the eigenspace corresponding to λ_0 is called **geometric multiplicity** of λ_0

The number of times that $\lambda - \lambda_0$ appears as a factor in the characteristic polynomial of A is called the **algebraic multiplicity** of λ_0

$$(\lambda - 1)(\lambda - 2)^2 = 0$$

algebraic multiplicity of $\lambda = 1$ is

algebraic multiplicity of $\lambda = 2$ is

Section 5.2 Diagonalization

THEOREM 5.25 Geometric and Algebraic Multiplicity

If A is a square matrix, then:

- a) For every eigenvalue of A , the geometric multiplicity is less than or equal to the algebraic multiplicity.*
- b) A is diagonalizable if and only if the geometric multiplicity of every eigenvalue is equal to the algebraic multiplicity.*

Example 2

Homework

Sec 5.2

True/False

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